

YANFU LIU

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🎓 Education

University of Science and Technology Beijing

Sep. 2023 – Expected Jun. 2027

B.E. in Artificial Intelligence; transferred from Bioengineering

🔍 Research Interests

Cloud-Edge AI: PEFT, distributed optimization, and incentive design.

AI for Biology: Biological imaging, molecular quantification, and controllable peptide design.

Agentic AI: Retrieval, tool use, planning, and multi-agent coordination.

🏢 Research Experience

Cloud-Edge Collaborative LLM Intelligence

Dec. 2024 – Present

City University of Hong Kong | Advisor: Dr. Huiming Chen

- **JANUS: Originated the reciprocal dual-phase framework** combining distributed LLM training, PEFT, and interleaved reconstruction. Established baselines and evaluation/ablation pipelines, tuned hyperparameters, and led the ACM SIGKDD R&R manuscript.
- **Cloud-Edge LLM Systems Survey: As corresponding author, proposed a taxonomy** of training and inference frameworks across cloud-edge architectures, conducted an extensive literature review, and led key manuscript sections connecting algorithmic theory with vertical applications.
- **Collaborative Economics and Incentive Design:** Refined the core GHM theoretical framework and translated its closed-form solutions into deployable evaluation metrics for end-to-end testbeds and simulations. Led the paper structure, literature synthesis, and full-draft development for the planned WWW submission.

AMPDesign: Interpretable Antimicrobial Peptide Design (AI for Biology)

Apr. 2026 – Present

The Chinese University of Hong Kong | Advisor: Prof. Yu Li

- **Multi-Objective Formulation:** Formulated antimicrobial peptide discovery as a problem balancing activity, hemolysis, cytotoxicity, selectivity, stability, and solubility.
- **Interpretable Representation:** Designed an ESM-2 pipeline with sparse autoencoders and concept bottlenecks to align latent factors with biological and physicochemical properties.
- **Controllability Validation:** Designed latent ablation, clamping, and sequence-editing experiments to validate activity–toxicity trade-offs.

CTransformer: 4D Cell Imaging and Molecular Quantification (CV for Biology)

Apr. – Jun. 2026

City University of Hong Kong | Advisor: Prof. Hong Yan

- **Segmentation and Tracking Evaluation:** Built evaluation setups for transformer-based 3D cell-membrane segmentation and lineage tracking, with performance comparisons and error analysis across developmental stages.
- **Molecular Quantification:** Connected reconstructed morphology and cell-contact context with molecular readouts, contributing validation evidence for a *Nature Methods* manuscript under revision.

🏗️ Engineering Projects

Industrial Digital Twin and Multi-Agent Decision Support

Oct. 2025 – Mar. 2026

City University of Hong Kong (CityU) | *Industry Partner* | *Collaborative R&D Project*

- **Digital-Twin Validation:** Translated wastewater-treatment requirements into evaluation criteria and designed test pipelines for the physical-process fidelity of a water–energy–carbon simulator.

- **Multi-Agent Decision Support:** Led agent-framework tuning and scenario design to evaluate operational trade-offs across treatment performance, energy consumption, and carbon emissions.
- **Enterprise RAG Integration:** Optimized retrieval and grounding for operational knowledge, and translated evaluation findings into recommendations for domain experts and enterprise stakeholders.

AI-Enabled Software Engineering Course PlatformMar. – Jun. 2026

University of Science and Technology Beijing (USTB) | *Project Lead* | *Campus Teaching Platform*

- **Agent Service Architecture:** Designed an agent-service layer for course-context Q&A, FAQ generation, and teaching support, orchestrating role-aware context routing and structured agent calls across course and assignment workflows through REST APIs and shared data models.
- **Campus Delivery and Leadership:** Led architecture and integration across five student teams to deliver the platform for real instructional use at USTB, unifying classes, courseware, assignments, notifications, analytics, JWT access control, and OSS storage.

 Working Papers

** denotes equal contribution, † denotes advisor.*

- **Yanfu Liu***, Xuebin Tang*, Huiming Chen†, Zelin Li, and Hong YAN. JANUS: Reciprocal Dual-Phase Learning for LLM Empowered Cloud-Edge Intelligence. *Revise and Resubmit (R&R), ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*.
- Huiming Chen†, **Yanfu Liu(corresponding author)**, Xuebin Tang, Hairui Yu, Zelin Li, and Hong YAN. Large Language Models Empowered Cloud-Edge Collaborative Intelligence: Frameworks, Optimizations and Applications. *Submitted to ACM Computing Surveys, Mar. 2026.*
- Xuebin Tang*, **Yanfu Liu***, Huiming Chen†, and Hong YAN. Invisible Chains, Visible Gains: Hold-Up and Self-Enforcing Cooperation in Cloud-Edge Intelligence. *Planned submission to The Web Conference (WWW), Aug. 2026.*
- Zelin Li, Guoye Guan, Xian Xiu, Dongying Xie, Yiming Ma, Chenwei Wang, **Yanfu Liu**, et al. CTransformer: Deep-transformer-based 3D Cell Membrane Tracking with Subcellular-resolved Molecular Quantification. *Submitted to Nature Methods; under revision.*
- Huiming Chen, **Yanfu Liu**, Xuebin Tang, Zhaobin Wei, Haoqiang Liu, Hong YAN. Robustifying Second-Order Accelerated Federated Learning: Algorithmic Design and Analysis. *Submitted to IEEE Transactions on Neural Networks and Learning Systems, under major revision.*

 Awards and Honors

National and International Awards

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| • ASC World Supercomputer Student Challenge, International Second Prize | 2025 |
| • National College Students Computer Games Competition, National Second Prize | 2025 |

Provincial, Ministerial and Professional Honors

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| • 2024 Beijing Engineering Practice and Innovation Ability Competition and 2025 National Qualifier (Future Technology Exploration Track), Grand Prize | 2025 |
| • CUMCM (Mathematical Contest in Modeling) Beijing Division, Second Prize | 2024 |
| • Computer Software Copyright, "Smart Boyue APP V1.0" | 2023 |

 Skills

Languages: English (CET6 passed, IELTS in preparation), Mandarin (Native)
Programming Languages: Python, LaTeX, C++, MATLAB
Tools and Platforms: Linux, Git, AI-Assisted Development
Database Technologies: SQLite
Frameworks and Libraries: PyTorch, HuggingFace Transformers, Pandas, NumPy.